DNS Flag Day...

are you prepared?

Sebastián Castro
NZNOG 2019
Napier

InternetNZ

Outline

- What is EDNS?
- How is EDNS used?
- What's the problem?
- What's DNS Flag Day?
- How the effects are measured?
- Measurements and results
- Some stories from the process

What is EDNS?

RFC 2671 -- 1999

RFC 6891 -- 2013

- Defines a backward compatible mechanism to signal support for new DNS options
- Original specification includes support for DNS responses larger than 512 bytes, extended response codes, etc.

How is EDNS used?

Current extensions:

- NSID -- RFC 5001, nameserver identification string
- DNSSEC -- DO bit, signals supports or interest for DNSSEC-related records
- Client-subnet, RFC 7871, signals the network the query comes from
- Keep-alive, RFC 7828, variable timeouts for DNS over TCP
- Cookies, IETF Draft, lightweight security mechanism
- and more to come

What's the problem?

- Authoritative DNS servers block responses, don't answer, or answer with the wrong packet.
  - In general, bad implementations of DNS not following the standards
- Poorly implemented firewalls on the way, poor firewall rules blocking valid traffic or unaware of the standards
- Resolvers have to send a query, wait for a timeout and retry using a different method: TCP or discard EDNS
  - Forces delays and thwarts innovation and deployment of new features

What's DNS Flag Day?

DNS implementations decided to remove workarounds in a coordinated way

- BIND, Unbound, PowerDNS and Knot will release new versions with the workarounds removed
- Support from Google DNS, Cloudflare, Quad 9, Cisco, Facebook

Feel the pain

- If you run inadequate DNS infrastructure, your domains will break

Why should I be concerned?

Measuring DNS flag day effects

“DNS Compliance Testing” tool written by ISC
https://gitlab.isc.org/isc-projects/DNS-Compliance-Testing

- Only check for EDNS compliance at this stage
- Same test available at https://dnsflagday.net

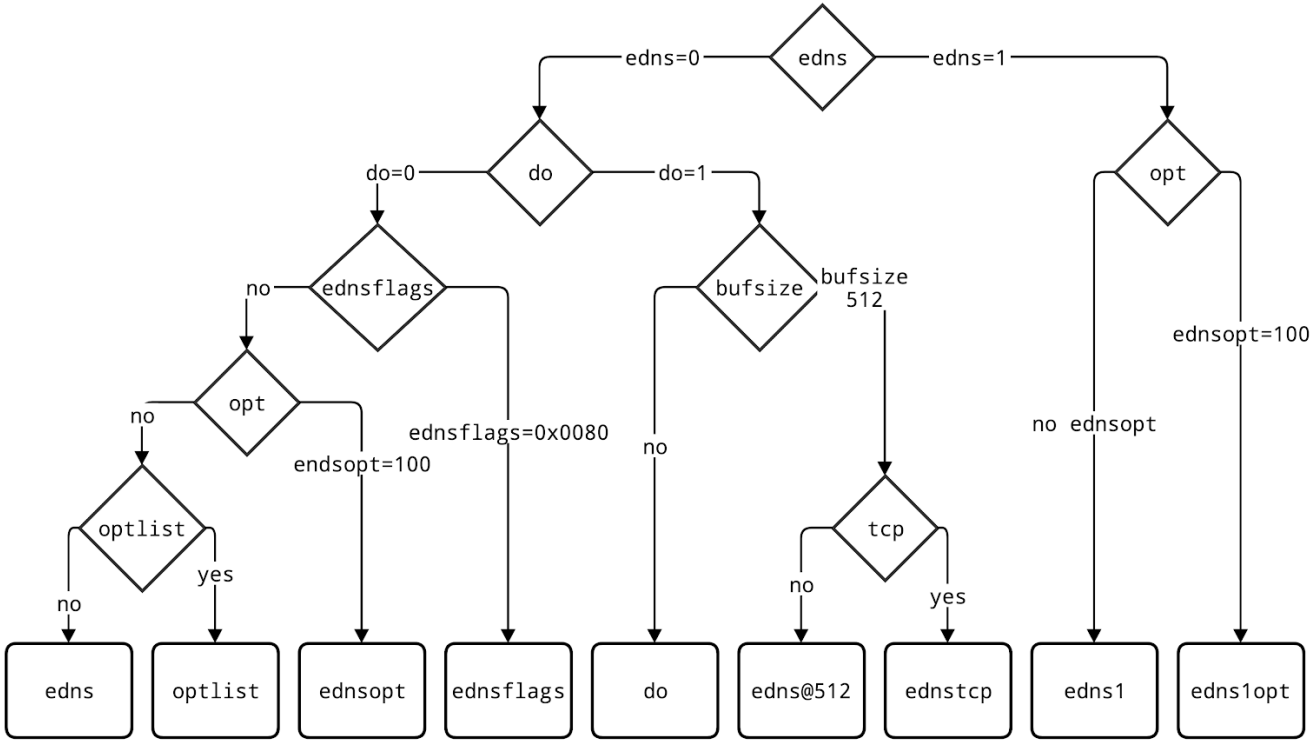
“EDNS Compliance scanner for DNS zones” from CZ.NIC:

- https://gitlab.labs.nic.cz/knot/edns-zone-scanner/tree/master
- Uniquely test all addresses of a nameserver
- Preprocess a TLD zone and generate the minimal set of nameserver tests
- Test multiple times to discard transient errors

InternetNZ's involvement

- Early measurement of impact in coordination with CZ and CL
  - First collection in July 2018
- Presentations in different conferences
- Blog post
  - Kindly promoted by APNIC
- Bi-weekly data collection on all .nz domains starting in October
- Communication campaign
  - Registrants, Registrars, DNS Operators, popular domains

EDNS compliance test

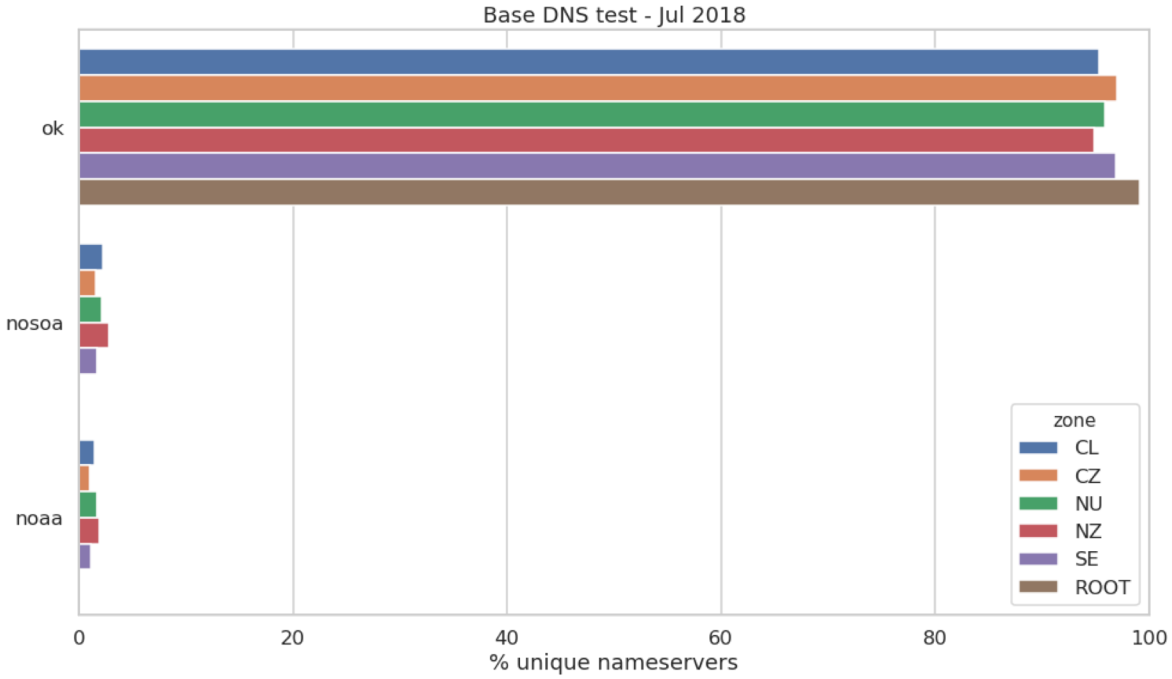


General statistics

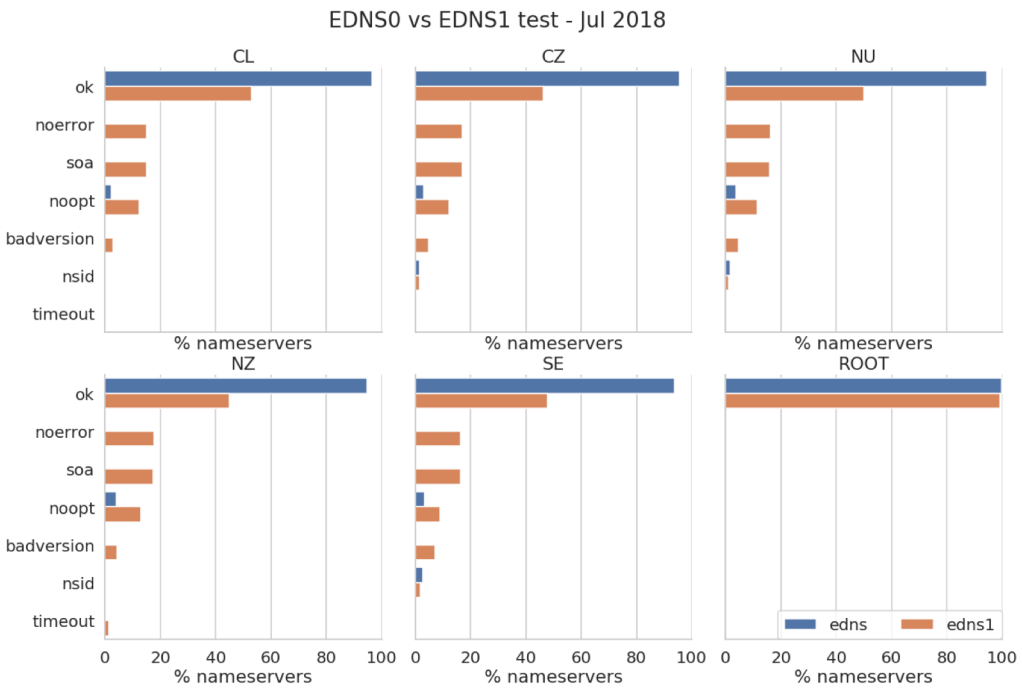
Date	# nameservers	# unique IPv4 addresses	# Unique IPv6 addresses	# Unique addresses
25 Jul 2018	21133	21219	4597	25816
25 Jan 2019	21360	21451	5214	26665

Comparing .nz with other ccTLDs

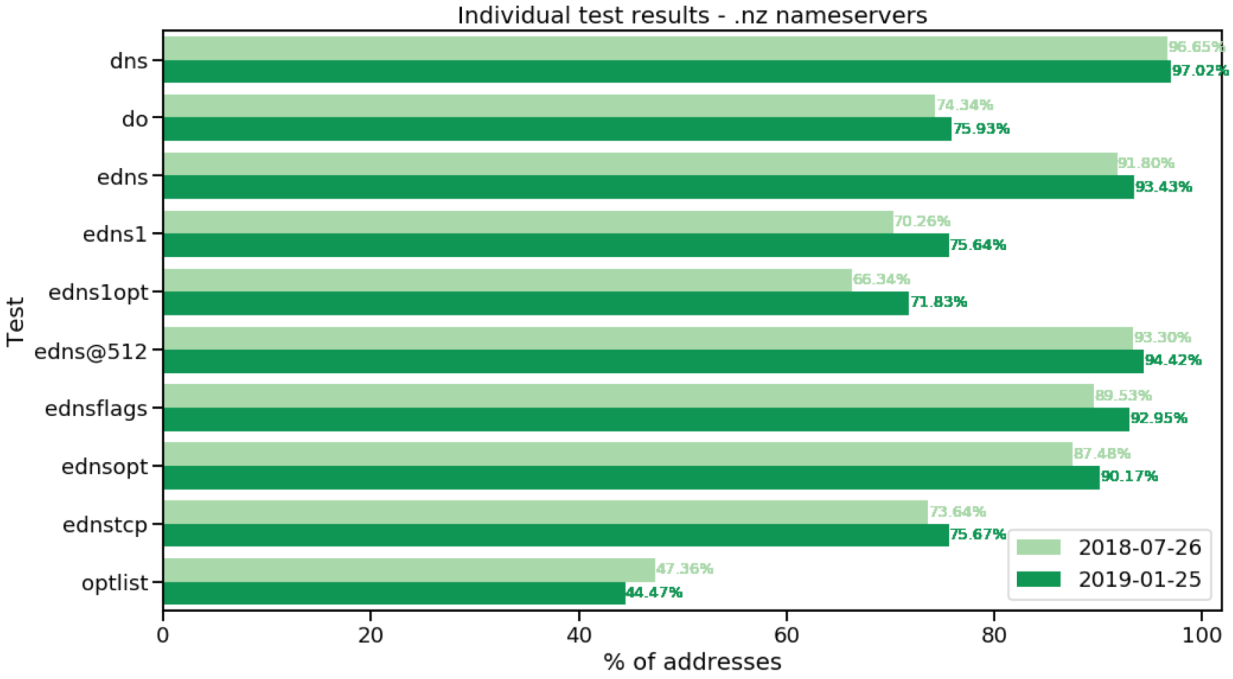
Five ccTLDs – base DNS test



Five ccTLDs – EDNS1 test



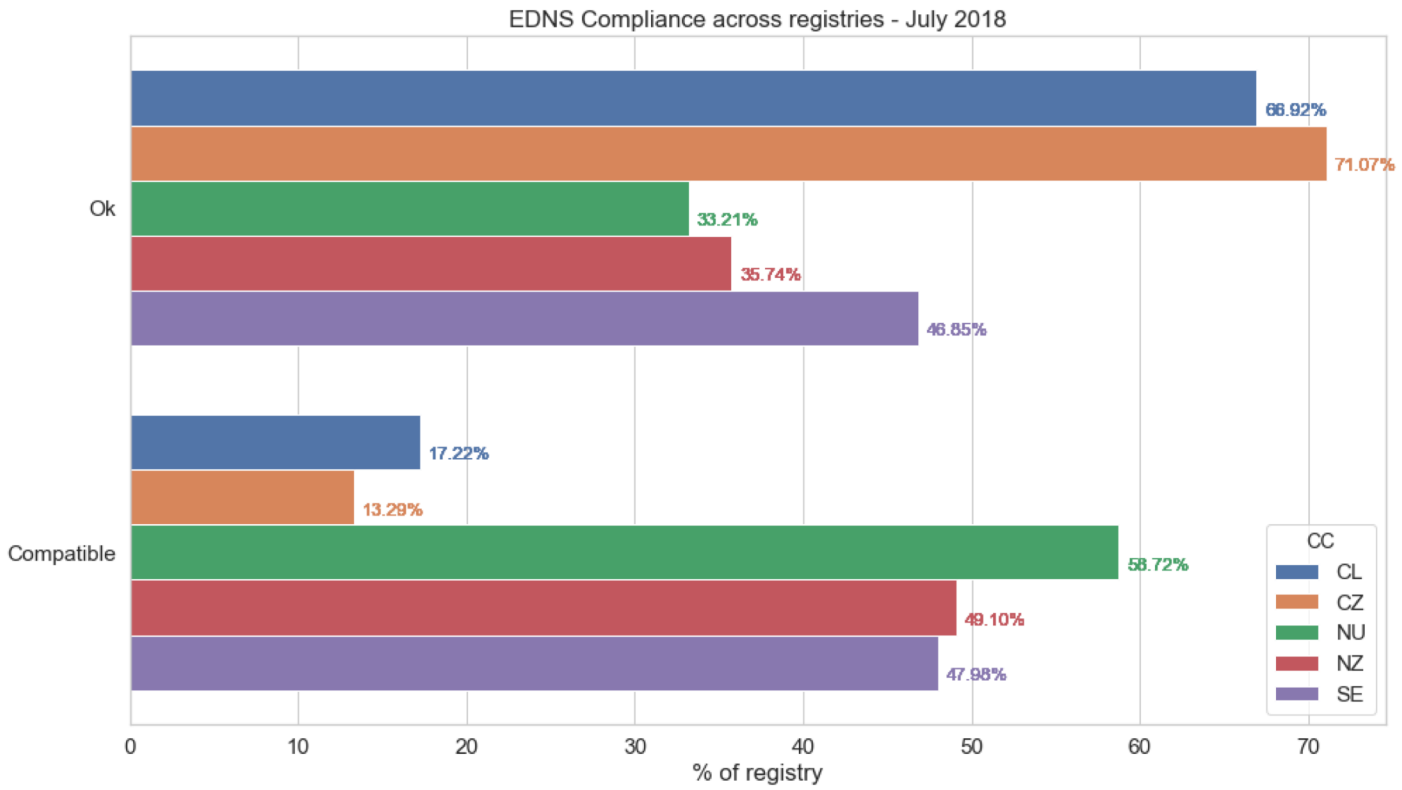
Compliance state by nameserver addresses



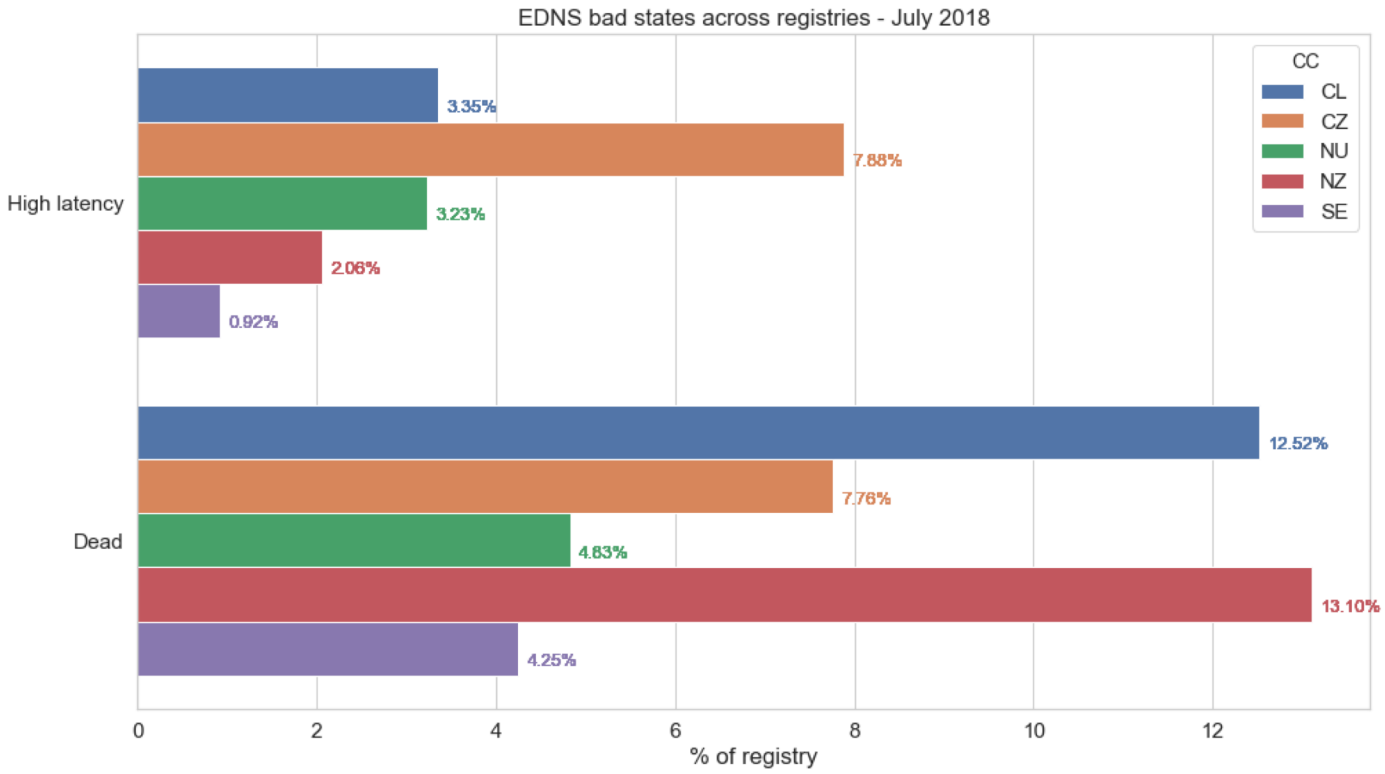
Domain status

- CZ.NIC created a tool that simplifies the understanding of the test
- Four states:
  - OK: All EDNS tests are ok
  - Compatible: None of the EDNS tests produce a timeout
  - High latency: Some nameserver addresses generate timeouts
  - Dead: All nameserver addresses generate timeouts
- Two set of rules
  - Permissive: Resolvers as they behave today
  - Strict: Resolvers with workarounds stripped

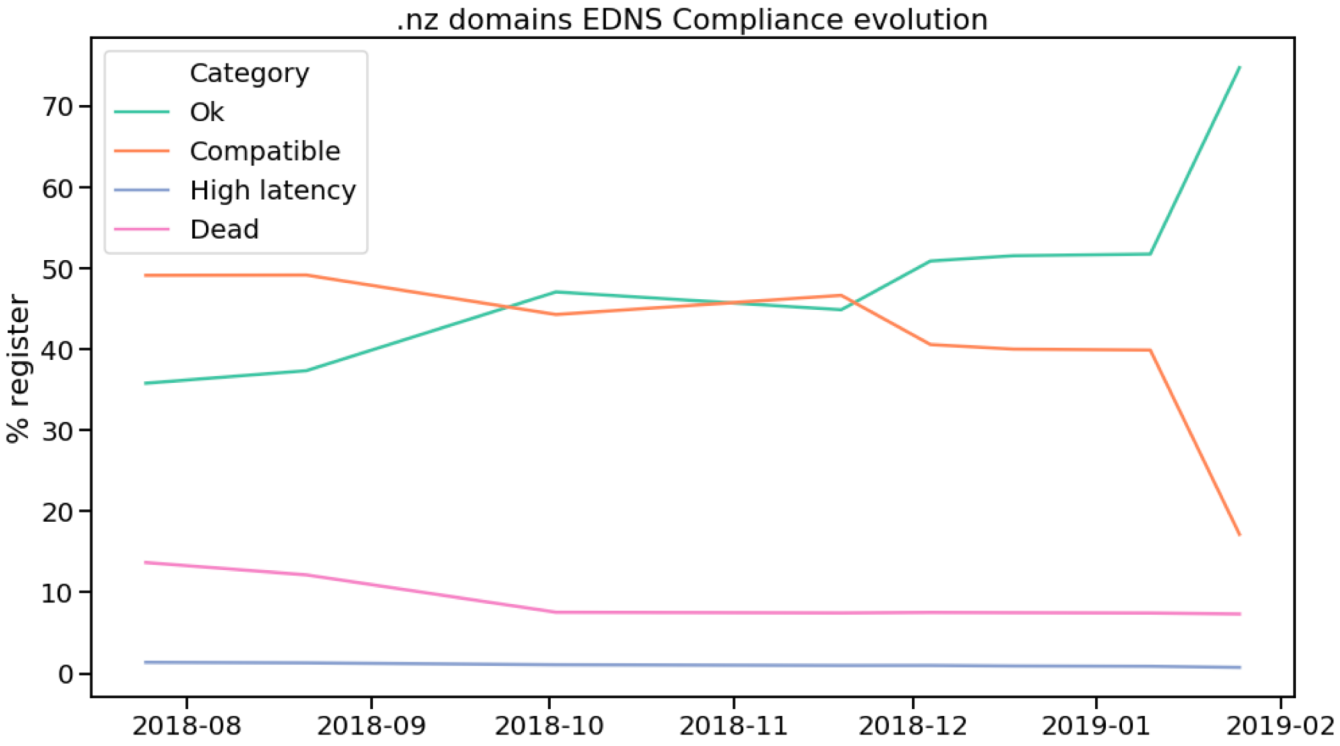
Five ccTLDs – domain status



Five ccTLDs – domain status



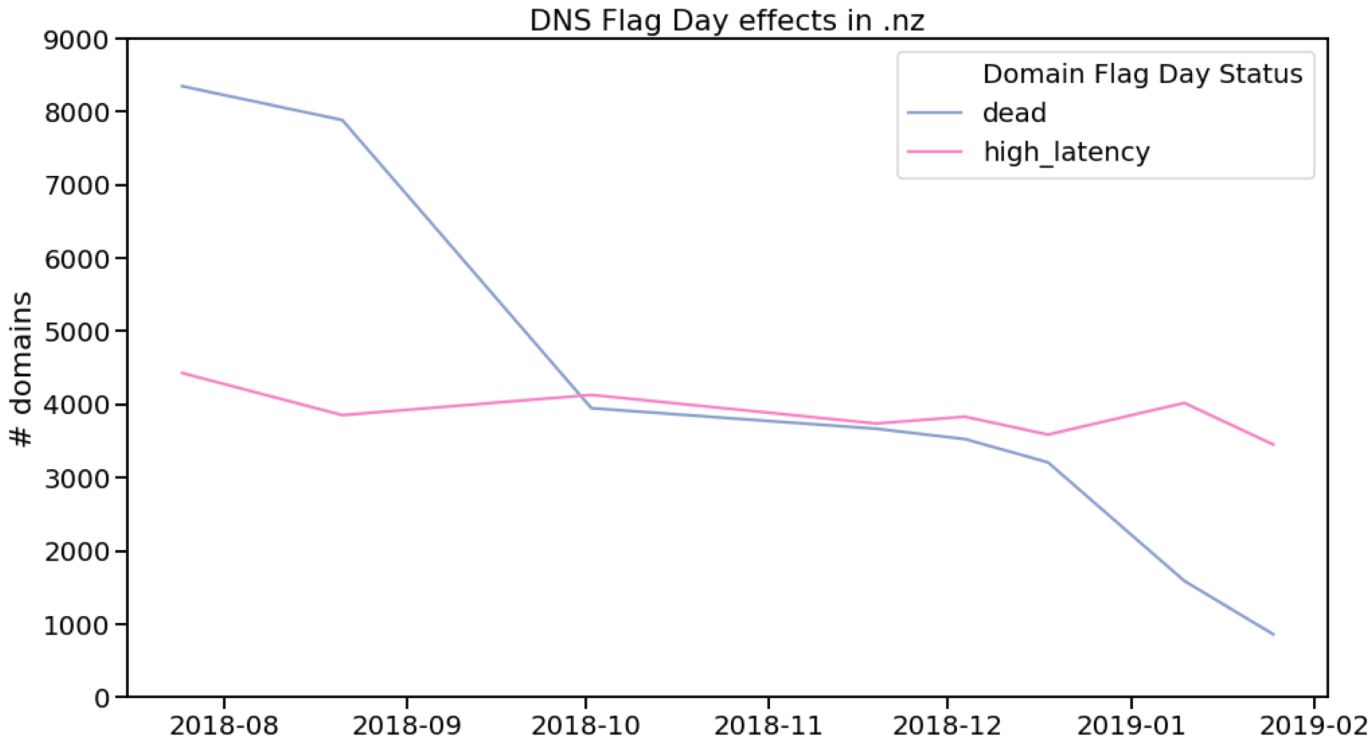
.nz improvements over time



Caveat

- Not all domains marked as high latency or dead are because EDNS brokenness
- There are a lot with lame nameservers, broken IP addresses, nameservers that can't resolve
- The reported domains differ presented a completely different picture!

Effectively affected .nz domains



How we got there?

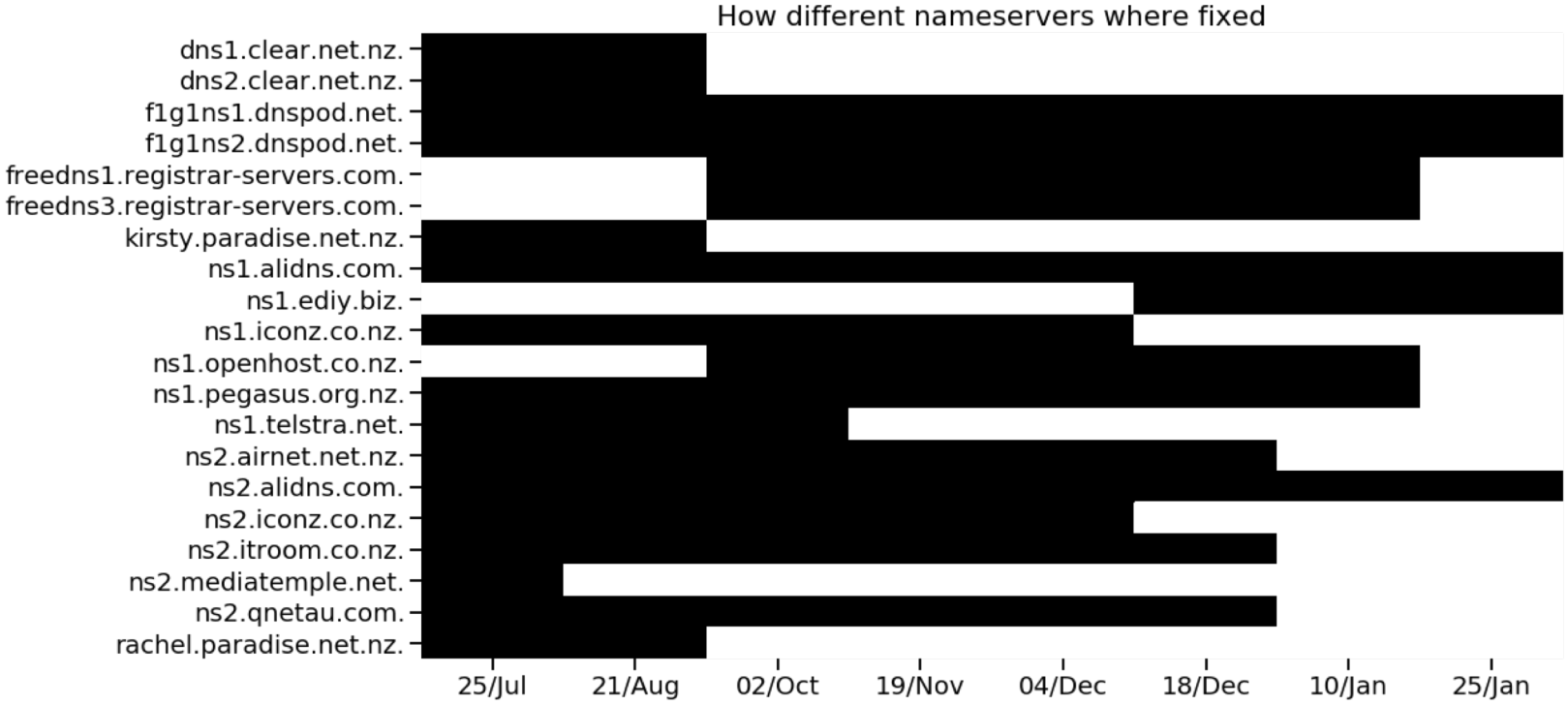
In the initial list of domains there were:

- A bunch of .govt.nz
- Banks
- Consultancy services
- Newspapers

We reached:

- Registrars
- DNS operators
- Registrants
- NZNOG via Slack

How we got there?



Some stories from the process

- Windows DNS server support
- Large DNS resolvers rollover plans
- Large authoritative DNS reaching out
- Some operators confused
- Some operators using old software
- Operators using modern software behind DPI firewalls
- “Why didn’t I find out about this sooner?”

Closing remarks

- It was hard to reach the right people in certain cases
- A big shout out to the team at InternetNZ and DNCL
  - Comms team, Channel Manager, Tech Services, Brent and his team
- We set the example for other ccTLDs to follow

¡GRACIAS!

sebastian@internetnz.net.nz